

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 22 — Analytical Techniques

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	22.1 Infrared Spectroscopy · 22.2 Mass Spectrometry · Characteristic IR Absorption Ranges · Molecular Ion
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	255 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

22.1 Infrared Spectroscopy

— Covalent bond vibrations (stretching and bending modes) and dipole moment change requirement

22.2 Mass Spectrometry: Molecular Ion & Carbon Count — Electron-impact ionisation to radical cations $[M]^+$; relative molecular mass M_r determination; $[M]$ **22.2 Mass Spectrometry: Isotope Patterns & Fragmentation** — Chlorine (^{35}Cl : ^{37}Cl approx 3:1) and bromine (^{79}Br : ^{81}Br approx 1:1) isotopic signatures in mono- and di-

SUBTOPIC 22.1 — INFRARED SPECTROSCOPY (PRINCIPLES, ABSORPTIONS & DIAGNOSTICS)

1 Principles of Infrared Spectroscopy & Bond Vibrations — 9701/22/M/J/23/Q5(a)–(d)

EASY · Syllabus 22.1

Infrared (IR) spectroscopy exploits the absorption of electromagnetic radiation by vibrating covalent bonds in molecules. Fig. 1.1 shows the characteristic infrared absorption ranges for key organic functional groups.

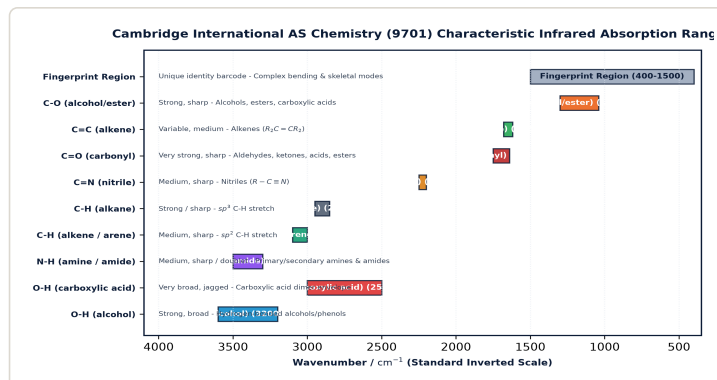


Fig. 1.1: Characteristic infrared absorption ranges for key organic bonds and functional groups.

((a)) State the physical condition that a molecular vibration must satisfy in order to absorb infrared radiation.

[1]

((b)) Explain why gaseous nitrogen, N₂, and oxygen, O₂, are transparent to infrared radiation, whereas carbon dioxide, CO₂, absorbs infrared radiation.

[2]

((c)) Identify the two primary types of vibrational modes that occur when covalent bonds absorb infrared radiation.

[2]

((d)) State the wavenumber range in cm⁻¹ associated with the fingerprint region of an infrared spectrum and explain its practical significance in chemical analysis.

[2]

2 Distinguishing Functional Group Profiles in IR — 9701/21/O/N/23/Q4(a)–(c)

HARD · Syllabus 22.1

Three constitutional isomers with molecular formula C₃H₆O, C₃H₈O, and C₃H₆O₂ were analyzed by infrared spectroscopy. Fig. 2.1 displays the recorded spectra profiles for propan-1-ol, propanoic acid, and propanal.

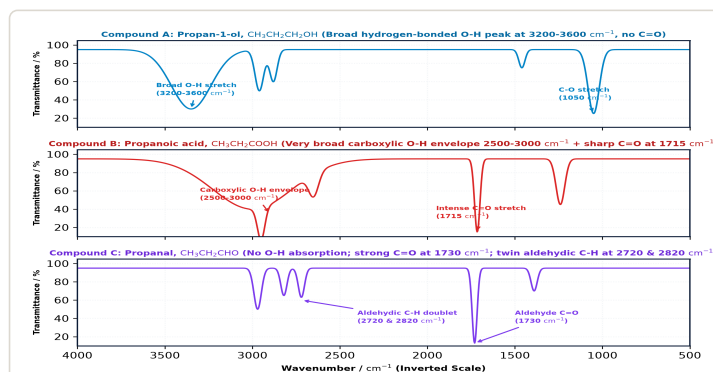


Fig. 2.1: Infrared spectra profiles for propan-1-ol, propanoic acid, and propanal.

- ((a))** Contrast the O–H absorption band observed in the spectrum of propan-1-ol with the O–H absorption band in propanoic acid, citing wavenumber ranges and peak appearances. **[3]**
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- ((b))** Identify the absorption peak present in both propanoic acid and propanal that is absent in propan-1-ol. Give its characteristic wavenumber range and the bond responsible. **[2]**
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- ((c))** Propanal exhibits a distinctive pair of sharp absorption bands at 2720 cm⁻¹ and 2820 cm⁻¹. Deduce the specific bond and group responsible for this doublet and explain how it differentiates propanal from propan-2-one. **[2]**
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3 Monitoring Oxidation of Primary Alcohols — 9701/22/F/M/24/Q3(a)–(d)

HARD · Syllabus 22.1

Ethanol was heated under reflux with excess acidified potassium dichromate(VI), K₂Cr₂O₇ / H₂SO₄, for 45 minutes. Samples of the reaction mixture were withdrawn at intervals and analyzed by infrared spectroscopy.

- ((a))** State the formula and color of the chromium species formed upon complete reduction of acidified dichromate(VI). **[1]**
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- ((b))** Describe the changes in the infrared spectrum as ethanol is oxidized initially to ethanal and subsequently to ethanoic acid. **[3]**
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- ((c))** Explain how infrared spectroscopy can prove that no unreacted ethanol remains in the purified reaction product. **[1]**
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- ((d))** State the apparatus modification required if the intended product was ethanal rather than ethanoic acid, and explain why this modification prevents complete oxidation. **[2]**
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4 Distinguishing Constitutional Isomers via Infrared Bands — 9701/23/M/J/23/Q3(a)–(c)

EASY · Syllabus 22.1

The two structural isomers X and Y both have the molecular formula C₂H₆O.

((a)) Draw the displayed formulas of isomers X and Y and state their systematic IUPAC names. [2]

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((b)) Explain how the infrared spectrum of isomer X can be distinguished from that of isomer Y by identifying key absorption bands present or absent in each. [2]

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((c)) State a simple chemical test with a positive observation that could confirm the identity of isomer X but would yield no reaction with isomer Y. [2]

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5 Infrared Analysis of Carbonyl vs Ester Isomers — 9701/22/O/N/22/Q4(a)–(c)

HARD · Syllabus 22.1

Compounds P and Q both possess the molecular formula $C_4H_8O_2$. Infrared spectroscopy shows that both compounds give a very strong absorption band at approximately 1735 cm^{-1} .

((a)) Compound P produces vigorous effervescence with aqueous sodium hydrogencarbonate, $NaHCO_3$, whereas compound Q does not react. [3]

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((b)) Describe the key difference between the infrared spectra of P and Q in the $2500\text{--}3600\text{ cm}^{-1}$ region. [2]

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((c)) Suggest the structure of compound Q given that alkaline hydrolysis followed by acidification yields methanol and propanoic acid. Name compound Q. [2]

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6 Infrared Spectra of Nitrogen-Containing Compounds — 9701/21/M/J/23/Q4(a)–(c)

HARD · Syllabus 22.1

Compounds J, K, and L are nitrogen-containing organic molecules with three carbon atoms: propanenitrile (CH_3CH_2CN), propylamine ($CH_3CH_2CH_2NH_2$), and propanamide ($CH_3CH_2CONH_2$).

((a)) Identify the characteristic absorption band that confirms the presence of the functional group in propanenitrile. State its wavenumber and bond assignment. [2]

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- ((b))** Describe the appearance and wavenumber range of the N-H absorption in the infrared spectrum of propylamine. Explain why primary amines typically exhibit a doublet. **[2]**
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- ((c))** State two characteristic absorption bands in the infrared spectrum of propanamide that together distinguish it from both propanenitrile and propylamine. **[2]**
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7**Monitoring Nucleophilic Addition to Carbonyls via IR — 9701/22/M/J/22/Q5(a)–(c)****HARD · Syllabus 22.1**

Ethanal reacts with hydrogen cyanide, HCN, in the presence of trace NaCN catalyst to produce 2-hydroxypropanenitrile.

- ((a))** State the role of the NaCN catalyst in terms of the reaction mechanism. **[1]**
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- ((b))** Describe how infrared spectroscopy can be used to monitor the progress of this reaction from start to finish. **[3]**
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- ((c))** Explain whether the product 2-hydroxypropanenitrile exhibits optical isomerism. Draw three-dimensional representations of any optical isomers formed. **[3]**
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8**Infrared Identification of Halogenoalkane Elimination — 9701/21/O/N/22/Q3(a)–(c)****EASY · Syllabus 22.1**

2-Bromopropane is heated under reflux with ethanolic potassium hydroxide, forming a gaseous organic compound W.

- ((a))** State the type of reaction taking place and identify organic product W. **[2]**
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- ((b))** Identify two significant new absorption bands in the infrared spectrum of W that are absent in the spectrum of 2-bromopropane. Give their wavenumber ranges and bond assignments. **[2]**
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- ((c))** Identify the absorption band present in 2-bromopropane that disappears when W is formed. **[1]**
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9 Distinguishing Aldehydes and Ketones by IR — 9701/22/F/M/23/Q4(a)–(c)

HARD · Syllabus 22.1

Pentan-2-one and pentanal are structural isomers with the molecular formula $C_5H_{10}O$. Both exhibit an intense absorption band at approximately 1720 cm^{-1} .

((a)) State the functional group type of pentan-2-one and pentanal and the bond responsible for the peak at 1720 cm^{-1} . **[2]**

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((b)) Explain in detail how the infrared spectrum of pentanal can be conclusively distinguished from that of pentan-2-one without using chemical reagents. **[2]**

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((c)) State a diagnostic chemical test that gives a visible positive result with pentan-2-one but a negative result with pentanal. Give reagents and the observation. **[2]**

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10 Greenhouse Gases & Atmospheric Infrared Absorption — 9701/22/M/J/21/Q2(a)–(c)

EASY · Syllabus 22.1

Incoming solar ultraviolet and visible radiation warms the Earth's surface, which re-emits electromagnetic radiation in the infrared region. Certain atmospheric gases absorb this infrared radiation, contributing to the greenhouse effect.

((a)) Name two major naturally occurring atmospheric gases that absorb infrared radiation and two major gases that do not. **[2]**

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((b)) Explain on a molecular level why carbon dioxide absorbs infrared radiation, referring to molecular geometry and dipole moments. **[2]**

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((c)) Methane has a significantly higher global warming potential (GWP) than carbon dioxide. Suggest two reasons why methane is a more potent greenhouse gas per molecule. **[2]**

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11 Hydrogen Bonding Effect on Infrared Absorption — 9701/23/O/N/23/Q3(a)–(c)

HARD · Syllabus 22.1

The infrared spectrum of liquid ethanol differs markedly from that of gaseous ethanol recorded at low vapor pressure.

- ((a))** In gaseous ethanol, the O–H absorption appears as a sharp peak at 3650 cm^{-1} , whereas in liquid ethanol it appears as a very broad band centered at 3350 cm^{-1} . Explain this difference in terms of intermolecular forces. **[3]**
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- ((b))** Explain why the broad absorption band shifts to a lower wavenumber when hydrogen bonding occurs. **[2]**
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- ((c))** State whether the C–H stretching band in ethanol is significantly broadened between the gas and liquid phases. Justify your answer. **[1]**
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12 Distinguishing Alcohols of Different Classes — 9701/21/M/J/22/Q3(a)–(c)

HARD · Syllabus 22.1

Compounds A, B, and C are isomeric alcohols with the formula $\text{C}_4\text{H}_{10}\text{O}$. Alcohol A is butan-1-ol, B is butan-2-ol, and C is 2-methylpropan-2-ol.

- ((a))** Classify alcohols A, B, and C as primary, secondary, or tertiary. **[1]**
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- ((b))** Explain why infrared spectroscopy alone cannot readily distinguish between alcohols A, B, and C based solely on functional group absorptions above 1500 cm^{-1} . **[2]**
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- ((c))** Describe how infrared spectroscopy can be combined with chemical oxidation to distinguish clearly between all three isomers. **[3]**
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13 Ester Hydrolysis Monitored by Infrared Spectroscopy — 9701/22/O/N/23/Q5(a)–(c)

EASY · Syllabus 22.1

Methyl ethanoate, $\text{CH}_3\text{COOCH}_3$, is heated under reflux with dilute sulfuric acid, establishing a dynamic chemical equilibrium.

- ((a))** Write the balanced chemical equation for the acid-catalyzed hydrolysis of methyl ethanoate. **[1]**
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- ((b))** Describe the infrared spectrum of pure methyl ethanoate before reaction, stating the two principal absorption bands. **[2]**
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- ((c)) Identify the new absorption bands that appear in the infrared spectrum as equilibrium is established, assigning each band to its corresponding product. [2]
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14 **Deducing Unknown Compound Structure from IR Data — 9701/21/F/M/22/Q4(a)–(d)**

HARD · Syllabus 22.1

A pure colorless liquid, compound D, has the empirical formula C_2H_4O and a relative molecular mass of 88.

- ((a)) Deduce the molecular formula of compound D. [1]
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- ((b)) The infrared spectrum of D exhibits strong absorptions at 1740 cm^{-1} and 1240 cm^{-1} , but no absorption between 2500 and 3600 cm^{-1} . Deduce the functional group present in D. [2]
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- ((c)) Compound D does not react with sodium metal and does not react with acidified potassium dichromate(VI). When heated with aqueous sodium hydroxide, it forms ethanol and a sodium salt of a carboxylic acid. Deduce the structural formula and name of D. [2]
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- ((d)) Write the equation for the saponification reaction of D with aqueous sodium hydroxide. [1]
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15 **Alkene Hydration & Polymerisation IR Diagnostics — 9701/22/M/J/24/Q4(a)–(c)**

HARD · Syllabus 22.1

Ethene can be converted into ethanol via catalytic hydration, or polymerised to poly(ethene).

- ((a)) State the reagents and industrial conditions used to convert ethene into ethanol. [2]
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- ((b)) Explain how infrared spectroscopy can demonstrate that ethene has completely reacted during industrial hydration to ethanol. [2]
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- ((c)) Describe the infrared spectrum of pure poly(ethene) and explain why it displays far fewer diagnostic absorption bands than ethene. [2]
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16 MCQ: Identifying Characteristic Wavenumber — 9701/12/M/J/23/Q38

EASY · Syllabus 22.1

An organic compound shows a very strong, sharp absorption band at 1715 cm^{-1} and a broad absorption band extending from 2500 cm^{-1} to 3000 cm^{-1} in its infrared spectrum.

((a))

Which compound is responsible for this infrared spectrum?

- A: Propan-1-ol
- B: Propanoic acid
- C: Ethyl methanoate
- D: Propan-2-one

[1]**17 MCQ: IR Transparency of Atmospheric Gases — 9701/11/O/N/23/Q37**

EASY · Syllabus 22.1

Which atmospheric gas does NOT absorb infrared radiation?

((a))

Select the gas that is transparent to infrared radiation.

- A: Carbon dioxide, CO_2
- B: Water vapor, H_2O
- C: Oxygen, O_2
- D: Methane, CH_4

[1]**18 MCQ: Functional Group Assignment at 2250 cm^{-1} — 9701/12/F/M/24/Q39**

EASY · Syllabus 22.1

An unknown compound exhibits a sharp absorption peak at 2235 cm^{-1} and no absorption between 1600 cm^{-1} and 1800 cm^{-1} .

((a))

Which functional group is present in this compound?

- A: Carbonyl, $\text{C}=\text{O}$
- B: Nitrile, $\text{C}\equiv\text{N}$
- C: Alkene, $\text{C}=\text{C}$
- D: Alcohol, $\text{O}-\text{H}$

[1]**19 MCQ: Monitoring Reaction Progress by IR — 9701/13/M/J/22/Q38**

HARD · Syllabus 22.1

Propan-2-one is treated with sodium borohydride, NaBH_4 , in aqueous ethanol.

((a))

Which statement correctly describes the change in the infrared spectrum as the reaction proceeds to completion?

- A: A peak at 1715 cm^{-1} appears and a broad peak at 3350 cm^{-1} disappears
- B: A peak at 1715 cm^{-1} disappears and a broad peak at 3350 cm^{-1} appears
- C: A sharp peak at 2250 cm^{-1} disappears and a peak at 1650 cm^{-1} appears
- D: A peak at 1240 cm^{-1} disappears and a broad peak at 2500 cm^{-1} appears

[1]**20 MCQ: Distinguishing Esters from Carboxylic Acids — 9701/11/M/J/23/Q39**

EASY · Syllabus 22.1

Which absorption band is present in ethanoic acid but completely absent in methyl methanoate?

- ((a)) Identify the diagnostic absorption band.
- A: Strong absorption at 1720 cm⁻¹
 - B: Strong absorption at 1250 cm⁻¹
 - C: Very broad absorption at 2500–3000 cm⁻¹
 - D: Sharp absorption at 2950 cm⁻¹
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[1]

21 Infrared Analysis of Nitrile Hydrolysis — 9701/22/M/J/23/Q4(a)–(c)

HARD · Syllabus 22.1

Propanenitrile is heated under reflux with dilute hydrochloric acid.

- ((a)) Write the balanced equation for the acidic hydrolysis of propanenitrile, showing the organic product and inorganic byproduct.
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[1]

- ((b)) Detail two major changes observed in the infrared spectrum between 1600 cm⁻¹ and 3500 cm⁻¹ as propanenitrile is fully converted into the organic product.
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[2]

- ((c)) If propanenitrile were instead reduced using LiAlH₄ in dry ether, state how the infrared spectrum of the product would differ from that of the hydrolysis product.
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[2]

22 Distinguishing Alkene from Alkane by Infrared Spectroscopy — 9701/21/O/N/21/Q4(a)–(b)

EASY · Syllabus 22.1

Hexane and hex-1-ene are both colorless liquids at room temperature.

- ((a)) Describe how infrared spectroscopy can be used to distinguish between samples of hexane and hex-1-ene.
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[2]

- ((b)) State a simple test-tube reaction that confirms the presence of the alkene group, including reagents and observations.
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[2]

23 Infrared Identification of Dehydration Products — 9701/23/M/J/22/Q3(a)–(c)

HARD · Syllabus 22.1

Butan-2-ol is heated with concentrated sulfuric acid at 170 °C, producing an isomeric mixture of alkenes.

- ((a)) Name the three isomeric alkene products formed in this reaction and classify their relationship.
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[2]

- ((b))** Explain why infrared spectroscopy cannot easily distinguish between cis-but-2-ene and trans-but-2-ene in the 1500–4000 cm⁻¹ region. **[2]**
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- ((c))** Trans-but-2-ene has a center of symmetry and exhibits no infrared absorption for the C=C stretch. Explain this phenomenon. **[2]**
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24 Infrared Breathalyser Analysis — 9701/22/F/M/22/Q3(a)–(b)

EASY · Syllabus 22.1

Roadside infrared breathalysers measure the concentration of ethanol in exhaled breath by monitoring infrared absorption at a specific wavelength.

- ((a))** The instrument measures absorption at approximately 2950 cm⁻¹. Identify the covalent bond responsible for this absorption and state why water vapor in the breath does not interfere at this precise frequency. **[2]**
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- ((b))** Explain how the degree of infrared absorption is related to the ethanol concentration in the motorist's blood, stating the physical law governing this relationship. **[2]**
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25 Comprehensive Spectral Deduction: Compound X — 9701/21/M/J/24/Q4(a)–(c)

HARD · Syllabus 22.1

Compound X has the molecular formula C₃H₆O. Its infrared spectrum shows a very strong, sharp absorption band at 1720 cm⁻¹. It exhibits no absorption between 2500 and 3600 cm⁻¹, and no absorption between 2700 and 2850 cm⁻¹.

- ((a))** Deduce the structure of compound X. Justify your answer by eliminating alternative functional group isomers. **[3]**
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- ((b))** Compound X is treated with hydrogen cyanide and trace potassium cyanide. Draw the displayed formula of the organic product formed. **[1]**
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- ((c))** Predict the key changes in the infrared spectrum when compound X is converted to the product in (b). **[2]**
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SUBTOPIC 22.2 — MASS SPECTROMETRY (MOLECULAR ION, ISOTOPIC RATIOS & FRAGMENTATION)

26 Principles of Mass Spectrometry & Molecular Ion Peak — 9701/22/M/J/23/Q5(e)–(h)

HARD · Syllabus 22.2

Mass spectrometry is an analytical technique used to determine relative molecular masses and deduce chemical structures. Fig. 26.1 displays the mass spectrum of propan-2-one.

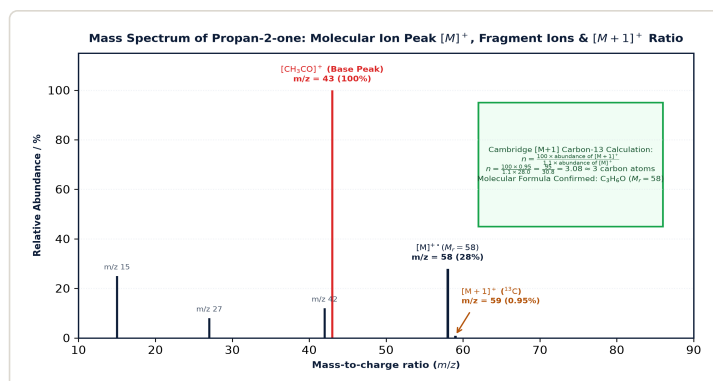


Fig. 26.1: Mass spectrum of propan-2-one showing molecular ion peak, fragment ions, and the $[M+1]^+$ isotope ratio.

- ((a)) Describe how gaseous molecules are ionised in an electron-impact mass spectrometer. Write an equation for the ionisation of propan-2-one, $(CH_3)_2CO$. [2]

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- ((b)) Explain what information is provided by the molecular ion peak, $[M]^+$, in a mass spectrum. [1]

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- ((c)) State the Cambridge formula used to calculate the number of carbon atoms, n , in an organic molecule from the relative abundances of the $[M]^+$ and $[M+1]^+$ peaks. [1]

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- ((d)) In the mass spectrum in Fig. 26.1, the $[M]^+$ peak at $m/z = 58$ has an abundance of 28.0% and the $[M+1]^+$ peak at $m/z = 59$ has an abundance of 0.95%. Calculate the number of carbon atoms, n , showing your working. [2]

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27 Chlorine & Bromine Isotope Signatures in Mass Spectrometry — 9701/21/O/N/23/Q5(a)–(d)

HARD · Syllabus 22.2

Naturally occurring chlorine consists of two stable isotopes, ^{35}Cl and ^{37}Cl , in an approximate 3:1 ratio. Bromine consists of two stable isotopes, ^{79}Br and ^{81}Br , in an approximate 1:1 ratio. Fig. 27.1 displays the isotopic multiplicity patterns for mono- and di-halogenated molecules.

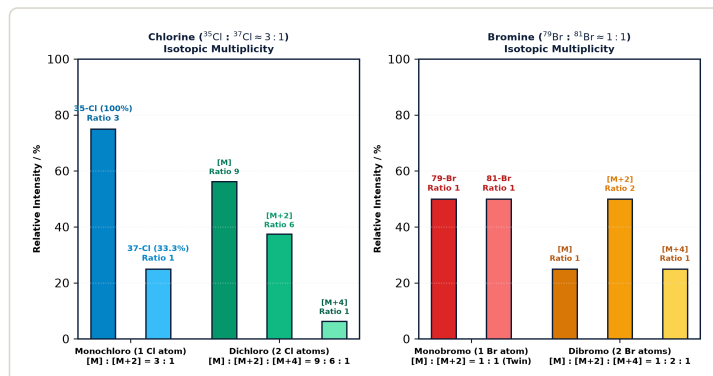


Fig. 27.1: Isotope peak patterns for compounds containing one or two chlorine or bromine atoms.

- ((a)) Describe the appearance of the molecular ion region for a compound containing a single chlorine atom, giving the ratio of the $[M]^+$ and $[M+2]^+$ peaks. [2]

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- ((b)) Describe the appearance of the molecular ion region for a compound containing a single bromine atom, giving the ratio of the $[M]^+$ and $[M+2]^+$ peaks. [2]

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- ((c)) A haloalkane containing two chlorine atoms displays three peaks in the molecular ion region: $[M]^+$, $[M+2]^+$, and $[M+4]^+$. Deduce the theoretical ratio of these three peak heights, showing mathematical justification. [3]

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- ((d)) State the ratio of the $[M]^+$, $[M+2]^+$, and $[M+4]^+$ peaks in a compound containing two bromine atoms. [1]

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28 Mixed Dihaloalkane Isotope Pattern — 9701/22/F/M/24/Q4(a)–(c)

HARD · Syllabus 22.2

1-Bromo-2-chloroethane, $\text{BrCH}_2\text{CH}_2\text{Cl}$, contains one chlorine atom and one bromine atom.

- ((a)) Calculate the relative molecular mass of 1-bromo-2-chloroethane containing the lightest isotopes (^{79}Br , ^{35}Cl , ^{12}C , ^1H). [1]

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- ((b)) Explain why the mass spectrum of 1-bromo-2-chloroethane exhibits three peaks in its molecular ion region at $m/z = 142$, 144, and 146. [2]

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- ((c)) Calculate the theoretical relative intensities of the peaks at $m/z = 142$, 144 , and 146 , expressing your answer as an integer ratio. [2]

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29 Fragmentation of Ketones & Base Peak Identification — 9701/22/M/J/22/Q4(a)–(c)

HARD · Syllabus 22.2

In the mass spectrum of propan-2-one (CH_3COCH_3), the base peak occurs at $m/z = 43$.

- ((a)) Define the term 'base peak' in mass spectrometry. [1]

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- ((b)) Identify the cation responsible for the base peak at $m/z = 43$ and write an equation showing its formation by fragmentation of the molecular ion. [2]

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- ((c)) A minor peak is observed at $m/z = 15$. Identify this species and explain why neutral radicals are not detected by mass spectrometers. [2]

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30 Carbon Count Calculation via $[M+1]^+$ Abundance — 9701/21/O/N/22/Q5(a)–(c)

HARD · Syllabus 22.2

An unknown hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass. The mass spectrum shows a molecular ion peak at $m/z = 56$ with a relative abundance of 38.5% and an $[M+1]^+$ peak at $m/z = 57$ with a relative abundance of 1.70%.

- ((a)) Calculate the empirical formula of the hydrocarbon. [2]

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- ((b)) Calculate the number of carbon atoms, n , in the molecule using the $[M+1]^+$ peak data. [2]

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- ((c)) Deduce the molecular formula of the hydrocarbon and suggest two possible structural isomers that are alkenes. [2]

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31 Fragmentation Differences in Isomeric Alcohols — 9701/22/M/J/21/Q4(a)–(c)

HARD · Syllabus 22.2

Pentan-2-ol and pentan-3-ol are structural isomers with the formula $\text{C}_5\text{H}_{12}\text{O}$ ($M_r = 88$).

((a)) Both isomers undergo alpha-cleavage readily. Explain what is meant by alpha-cleavage in alcohols. [1]

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((b)) Pentan-3-ol, $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$, shows an intense fragment peak at $m/z = 59$. Identify the fragment cation and the neutral radical lost. [2]

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((c)) Pentan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$, can undergo two different alpha-cleavages, producing prominent fragment peaks at $m/z = 45$ and $m/z = 73$. Identify the fragment cation formed in each cleavage. [2]

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32 Ester Fragmentation Patterns — 9701/23/O/N/22/Q4(a)-(c)

HARD · Syllabus 22.2

The mass spectrum of methyl ethanoate, $\text{CH}_3\text{COOCH}_3$ ($M_r = 74$), exhibits major peaks at $m/z = 74, 59, 43, 31$, and 15.

((a)) Identify the ion responsible for the molecular ion peak at $m/z = 74$. [1]

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((b)) Identify the fragment cations responsible for the peaks at $m/z = 43$ and $m/z = 59$. [2]

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((c)) The peak at $m/z = 43$ corresponds to the loss of a neutral radical of mass 31. Identify this neutral species and write the fragmentation equation. [2]

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33 Distinguishing Aldehydes & Ketones by MS Fragmentation — 9701/21/M/J/23/Q5(a)-(c)

HARD · Syllabus 22.2

Propanal and propan-2-one both have the molecular formula $\text{C}_3\text{H}_6\text{O}$ ($M_r = 58$).

((a)) The mass spectrum of propanal shows a distinct peak at $m/z = 57$ ($M - 1$). Identify the species responsible and explain why ketones do not readily form an ($M - 1$) peak. [2]

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((b)) Explain why propan-2-one exhibits a very strong peak at $m/z = 43$, whereas propanal exhibits a strong peak at $m/z = 29$. [2]

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- ((c)) Identify the two possible cations that can give rise to a peak at $m/z = 29$ in the mass spectrum of propanal. [2]
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34 Characteristic Neutral Losses in Mass Spectrometry — 9701/22/F/M/23/Q5(a)–(d)

HARD · Syllabus 22.2

Fragment peaks often arise from the loss of common small neutral molecules or radicals from the molecular ion $[M]^+$.

- ((a)) Identify the neutral molecule lost when an alcohol undergoes dehydration in a mass spectrometer, producing an $(M - 18)$ peak. [1]
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- ((b)) Identify the neutral radical lost when an ethanoic acid ester produces an $(M - 15)$ peak. [1]
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- ((c)) Identify the neutral species of mass 28 lost when a primary alcohol or carbonyl compound fragments. [1]
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- ((d)) Carboxylic acids frequently exhibit an intense fragment peak corresponding to $(M - 17)$ and a peak at $m/z = 45$. Identify the fragments responsible for both observations. [2]
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35 Structural Elucidation of Alkyl Halide via MS — 9701/21/O/N/21/Q5(a)–(d)

HARD · Syllabus 22.2

An unknown bromoalkane B was analyzed by mass spectrometry.

- ((a)) The mass spectrum of B exhibits twin molecular ion peaks of equal height at $m/z = 136$ and $m/z = 138$. Deduce the molecular formula of B. [2]
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- ((b)) The base peak of B occurs at $m/z = 57$, corresponding to the loss of a bromine radical ($\cdot\text{Br}$). Deduce the formula of this base peak cation. [1]
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- ((c)) Compound B undergoes nucleophilic substitution with aqueous sodium hydroxide exclusively via an $\text{S}_{\text{N}}1$ mechanism. Deduce the structural formula and systematic name of B. [2]
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((d))

Explain why compound B reacts via S_N1 rather than S_N2 .

[2]

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36

MCQ: Determining Number of Carbons from $[M+1]^+$ — 9701/12/M/J/23/Q39

EASY · Syllabus 22.2

In the mass spectrum of an organic compound, the $[M]^+$ peak at $m/z = 72$ has a relative abundance of 50.0% and the $[M+1]^+$ peak at $m/z = 73$ has a relative abundance of 2.20%.

((a))

How many carbon atoms are present in one molecule of this compound?

[1]

- A: 2
- B: 3
- C: 4
- D: 5

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37

MCQ: Halogen Identification from Isotopic Peak Heights — 9701/11/O/N/22/Q38

EASY · Syllabus 22.2

The mass spectrum of an organic compound contains molecular ion peaks at $m/z = 108$ and $m/z = 110$ with peak heights in the ratio 1 : 1.

((a))

Which halogen atom is present in this molecule?

[1]

- A: Fluorine
- B: Chlorine
- C: Bromine
- D: Iodine

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38

MCQ: Base Peak Identification for Propan-2-ol — 9701/12/F/M/23/Q38

HARD · Syllabus 22.2

When propan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$, is analyzed by mass spectrometry, the base peak is observed at $m/z = 45$.

((a))

Which species is responsible for this base peak?

[1]

- A: $[\text{CH}_3\text{CH}_2\text{O}]^+$
- B: $[\text{CH}_3\text{CH}=\text{OH}]^+$
- C: $[\text{COOH}]^+$
- D: $[\text{CH}_3\text{OCH}_2]^+$

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39

MCQ: Dichloroalkane Isotopic Multiplicity — 9701/13/M/J/23/Q39

HARD · Syllabus 22.2

Dichloromethane, CH_2Cl_2 , has molecular ion peaks at $m/z = 84$, 86, and 88.

((a))

What is the relative ratio of the peak heights at $m/z = 84$, 86, and 88?

[1]

- A: 1 : 2 : 1
- B: 3 : 2 : 1
- C: 9 : 6 : 1
- D: 3 : 1 : 1

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40 MCQ: Tropylium Ion Formation — 9701/11/M/J/22/Q39

HARD · Syllabus 22.2

The mass spectrum of an alkylbenzene exhibits an extremely stable base peak at $m/z = 91$.

((a))

Which ion corresponds to $m/z = 91$?

A: $[C_6H_5]^+$

B: $[C_6H_5CH_2]^+$ (tropylium cation, $[C_7H_7]^+$)

C: $[C_6H_5O]^+$

D: $[C_7H_9]^+$

[1]

41 Carbocation Stability in Mass Spectrometry Fragmentation — 9701/22/O/N/23/Q4(a)-(c)

HARD · Syllabus 22.2

The mass spectra of butane and 2-methylpropane (both C_4H_{10} , $M_r = 58$) display very different base peaks.

((a))

Explain why the base peak in 2-methylpropane, $CH(CH_3)_3$, is at $m/z = 43$ and has an abundance much greater than that in butane.

[2]

((b))

Write the equation for the formation of the $m/z = 43$ carbocation from the 2-methylpropane molecular ion.

[1]

((c))

State the relative stabilities of primary, secondary, and tertiary carbocations and explain the electronic factor responsible.

[2]

42 Molecular Formula Deduction from Elemental Analysis & MS — 9701/21/M/J/22/Q5(a)-(c)

HARD · Syllabus 22.2

An organic compound G was subjected to elemental combustion analysis and mass spectrometry. Combustion of 1.32 g of G produced 2.64 g of CO_2 and 1.08 g of H_2O .

((a))

Calculate the masses of carbon, hydrogen, and oxygen present in the 1.32 g sample of G and determine its empirical formula.

[3]

((b))

In the mass spectrum of G, the molecular ion peak occurs at $m/z = 88$ with an intensity of 42.0%, and the $[M+1]^+$ peak occurs at $m/z = 89$ with an intensity of 1.85%. Determine the molecular formula of G.

[2]

((c))

The infrared spectrum of G shows strong absorptions at 1715 cm^{-1} and a very broad band from 2500 to 3000 cm^{-1} . Deduce the structural formula and IUPAC name of G.

[2]

43 Fragmentation of Primary Amines — 9701/22/M/J/24/Q5(a)–(c)

HARD · Syllabus 22.2

Propylamine, $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$, has a relative molecular mass of 59. Its mass spectrum shows a base peak at $m/z = 30$.

- ((a))** Explain how the base peak at $m/z = 30$ is formed by alpha-cleavage of the propylamine molecular ion. Identify the fragment cation and the radical lost. [2]

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- ((b))** Explain why the $[\text{CH}_2\text{NH}_2]^+$ fragment ion ($m/z = 30$) is exceptionally stable. [2]

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- ((c))** Predict the m/z of the base peak in the mass spectrum of the isomeric amine propan-2-amine, $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_3$, and identify the fragment ion. [2]

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44 Distinguishing Isomeric Ketones by MS — 9701/23/O/N/23/Q4(a)–(c)

HARD · Syllabus 22.2

Pentan-2-one ($\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$) and pentan-3-one ($\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$) are positional isomers with $M_r = 86$.

- ((a))** State the two distinct acylium fragment ions formed by alpha-cleavage of pentan-2-one, giving their m/z values. [2]

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- ((b))** State the single major acylium fragment ion formed by alpha-cleavage of symmetrical pentan-3-one, giving its m/z value. [2]

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- ((c))** Explain how mass spectrometry readily distinguishes between pentan-2-one and pentan-3-one based on these fragment peaks. [1]

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45 Combined Spectroscopic Elucidation: Unknown Ester — 9701/21/F/M/24/Q5(a)–(d)

HARD · Syllabus 22.2

Compound K is an ester with $M_r = 102$. Its mass spectrum shows a molecular ion peak at $m/z = 102$ (15%) and $[\text{M}+1]^+$ at $m/z = 103$ (0.83%). The base peak is at $m/z = 57$, and another prominent peak occurs at $m/z = 45$.

- ((a))** Calculate the number of carbon atoms, n , in compound K. [2]

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((b)) Determine the molecular formula of compound K. [1]

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((c)) The peak at $m/z = 57$ corresponds to $[C_2H_5CO]^+$ and the peak at $m/z = 45$ corresponds to $[OC_2H_5]^+$. Deduce the structural formula and IUPAC name of ester K. [2]

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((d)) Identify the two organic compounds formed when ester K is heated under reflux with aqueous sodium hydroxide followed by acidification. [2]

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46 Combined Deduction: Carbonyl Structural Isomers — 9701/22/M/J/23/Q3(a)–(d)

HARD · Syllabus 22.2

Compound M has $M_r = 72$. Elemental analysis reveals that M contains 66.7% carbon, 11.1% hydrogen, and 22.2% oxygen by mass. Its infrared spectrum exhibits a strong sharp absorption at 1718 cm^{-1} .

((a)) Show that the molecular formula of compound M is C_4H_8O . [2]

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((b)) Compound M reacts with 2,4-dinitrophenylhydrazine (2,4-DNPH) to form an orange precipitate, but produces no mirror or precipitate with Tollens' reagent. Deduce the class of organic compound and the structural formula of M. [2]

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((c)) The mass spectrum of M shows two major fragment peaks at $m/z = 43$ and $m/z = 57$. Identify the cations responsible for both peaks. [2]

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((d)) Write the structural equation for the reaction of compound M with alkaline aqueous iodine (the iodoform reaction). [2]

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47 Deducing Halogenoalkane Structure from MS Fragmentation — 9701/21/O/N/23/Q3(a)–(c)

HARD · Syllabus 22.2

Two isomeric chloroalkanes, X and Y, have the molecular formula C_3H_7Cl ($M_r = 78$ and 80 in 3:1 ratio).

((a)) State the displayed formulas and IUPAC names of isomers X and Y. [2]

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((b)) In the mass spectrum of X, the base peak occurs at $m/z = 43$. In the mass spectrum of Y, the base peak occurs at $m/z = 63$ and 65 (ratio 3:1). Explain how these base peaks allow X and Y to be unambiguously identified. **[3]**

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((c)) Identify the fragment ions at $m/z = 43$ and $m/z = 63$. **[2]**

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48 Synoptic Deduction: Alcohol vs Ether vs Carbonyl — 9701/22/F/M/22/Q5(a)–(d)

HARD • Syllabus 22.2

Four organic compounds E, F, G, and H are constitutional isomers with the molecular formula C_3H_8O .

((a)) Show that only three constitutional isomers actually exist for C_3H_8O and draw their skeletal structures. **[2]**

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((b)) Explain how infrared spectroscopy can immediately separate the three isomers into two distinct functional categories. **[2]**

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((c)) Mass spectrum of isomer 1 shows a base peak at $m/z = 31$ ($[CH_2OH]^+$). Mass spectrum of isomer 2 shows a base peak at $m/z = 45$ ($[CH_3CHOH]^+$). Deduce the identities of isomers 1 and 2. **[2]**

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((d)) The third isomer does not react with sodium and has no infrared peak above 3000 cm^{-1} . Name this isomer. **[1]**

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49 Synoptic Identification of Carboxylic Acid & Derivative — 9701/21/M/J/23/Q3(a)–(d)

HARD • Syllabus 22.2

Compound Z contains 40.0% carbon, 6.7% hydrogen, and 53.3% oxygen by mass. Its mass spectrum gives $[M]^+$ at $m/z = 60$ (25.0%) and $[M+1]^+$ at $m/z = 61$ (0.55%).

((a)) Calculate the empirical formula of compound Z. **[2]**

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((b)) Confirm the molecular formula of compound Z using the $[M+1]^+$ peak data. **[2]**

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((c)) Compound Z dissolves in water to form a solution with pH 3. Its infrared spectrum displays a broad band extending from 2500 cm^{-1} to 3000 cm^{-1} and a sharp peak at 1715 cm^{-1} . Deduce the structural formula of Z. [2]

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((d)) Compound Z is reacted with thionyl chloride, SOCl_2 . State the structural formula of the organic product and write the balanced equation. [2]

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Comprehensive Synoptic Spectral Elucidation — 9701/22/M/J/24/Q6(a)–(e)

HARD · Syllabus 22.2

Two isomeric organic compounds, S and T, have the molecular formula $\text{C}_5\text{H}_{10}\text{O}$ ($M_r = 86$). Both compounds give an orange precipitate when treated with 2,4-dinitrophenylhydrazine.

((a)) State the functional group type common to both compounds S and T based on the 2,4-DNPH test. [1]

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((b)) Compound S forms a silver mirror with Tollens' reagent. Its infrared spectrum shows a sharp carbonyl absorption at 1725 cm^{-1} and two distinctive sharp absorptions at 2720 cm^{-1} and 2820 cm^{-1} . Its mass spectrum has a base peak at $m/z = 57$. Deduce the displayed formula and IUPAC name of compound S, and identify the base peak cation. [3]

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((c)) Compound T does NOT react with Tollens' reagent. When treated with alkaline aqueous iodine, compound T forms a pale yellow precipitate of tri-iodomethane. Deduce the structural feature that compound T must contain. [1]

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((d)) The mass spectrum of compound T shows a base peak at $m/z = 43$ ($[\text{CH}_3\text{CO}]^+$) and an intense fragment peak at $m/z = 71$ ($[\text{M} - 15]^+$). Deduce the displayed formula and systematic name of compound T. [2]

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((e)) Compound T is reduced with NaBH_4 in aqueous ethanol, forming compound U. State the IUPAC name of compound U, and describe how its infrared spectrum differs from that of compound T. [2]

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Mark Scheme — Topic 22 — Analytical Techniques

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	There must be a change in the molecule's dipole moment during the vibration [1]	1
1(b)	N ₂ and O ₂ are homonuclear non-polar diatomic molecules that undergo no change in dipole moment when vibrating [1] CO ₂ undergoes asymmetrical stretching or bending vibrations that cause a net change in dipole moment, allowing IR absorption [1]	2
1(c)	Stretching vibrations (symmetrical or asymmetrical changes in bond length) [1] Bending vibrations (changes in bond angles / deformation) [1]	2
1(d)	Wavenumber range: Below 1500 cm ⁻¹ (or 400–1500 cm ⁻¹) [1] Significance: It contains a unique, complex pattern of bending and skeletal absorption bands characteristic of the whole molecule; matching this pattern against a reference spectral library confirms exact identity [1]	2
2(a)	Propan-1-ol exhibits a strong, broad absorption band at 3200–3600 cm ⁻¹ due to hydrogen-bonded alcohol O–H stretch [1] Propanoic acid exhibits a very broad, jagged absorption envelope spanning 2500–3000 cm ⁻¹ due to carboxylic acid dimer O–H stretch [1] The carboxylic acid O–H envelope overlaps the alkyl C–H stretching vibrations around 2950 cm ⁻¹ , whereas alcohol O–H does not overlap [1]	3
2(b)	Carbonyl group C=O stretch [1] Wavenumber range: 1640–1750 cm ⁻¹ (or 1680–1740 cm ⁻¹ for saturated carbonyls) [1]	2
2(c)	Aldehydic C–H stretching doublet (Fermi resonance) [1] Propan-2-one is a ketone lacking a hydrogen atom bonded directly to the carbonyl carbon, so it shows no absorptions at 2720/2820 cm ⁻¹ [1]	2
3(a)	Cr ³⁺ (or [Cr(H ₂ O) ₆] ³⁺), green solution [1]	1
3(b)	Ethanol to ethanal: disappearance of the broad alcohol O–H stretch at 3200–3600 cm ⁻¹ and appearance of sharp C=O at ~1730 cm ⁻¹ [1] Appearance of aldehydic C–H doublet at 2720 and 2820 cm ⁻¹ [1] Ethanal to ethanoic acid: appearance of very broad carboxylic acid O–H envelope at 2500–3000 cm ⁻¹ and shift of C=O to ~1710 cm ⁻¹ [1]	3
3(c)	Complete absence of the broad alcohol O–H absorption band at 3200–3600 cm ⁻¹ and absence of the C–O band at ~1050 cm ⁻¹ [1]	1
3(d)	Use simple distillation apparatus (distil immediately as formed) instead of heating under reflux [1] Ethanal has a lower boiling point (21 °C) due to lacking hydrogen bonding, so it evaporates and is condensed away from the oxidizing agent before it can oxidize further [1]	2
4(a)	Ethanol (CH ₃ CH ₂ OH) with all bonds shown [1] Methoxymethane (CH ₃ OCH ₃) with all bonds shown [1]	2
4(b)	Ethanol exhibits a strong, broad hydrogen-bonded O–H absorption at 3200–3600 cm ⁻¹ [1] Methoxymethane is an ether with no O–H bond, showing no absorption above 3000 cm ⁻¹ except sp ³ C–H stretch (2850–2950 cm ⁻¹) and C–O stretch at 1050–1300 cm ⁻¹ [1]	2
4(c)	Add a piece of sodium metal at room temperature [1] Ethanol produces effervescence / bubbles of hydrogen gas; methoxymethane shows no reaction [1]	2
5(a)	P is a carboxylic acid (contains –COOH group) because it reacts with NaHCO ₃ to liberate CO ₂ gas [1] Equation: RCOOH + NaHCO ₃ → RCOONa + H ₂ O + CO ₂ [1] P is butanoic acid (or 2-methylpropanoic acid) [1]	3
5(b)	Compound P shows a very broad, characteristic carboxylic acid O–H absorption envelope at 2500–3000 cm ⁻¹ [1] Compound Q shows no absorption in the 2500–3600 cm ⁻¹ region except for sharp sp ³ C–H stretches at 2850–2950 cm ⁻¹ [1]	2
5(c)	Structure: CH ₃ CH ₂ COOCH ₃ [1] Name: Methyl propanoate [1]	2
6(a)	Wavenumber: 2200–2250 cm ⁻¹ (sharp, medium intensity) [1] Bond: C≡N (nitrile group stretch) [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
6(b)	Wavenumber: 3300–3500 cm ⁻¹ [1] A primary amine has two N–H bonds that couple together to produce two distinct stretching modes: symmetric and asymmetric N–H stretch, resulting in a doublet (two peaks) [1]	2
6(c)	Strong carbonyl C=O stretch at 1640–1690 cm ⁻¹ (amide I band) [1] N–H stretching bands at 3300–3500 cm ⁻¹ (amide II / N–H stretch) [1]	2
7(a)	NaCN provides a high concentration of cyanide nucleophile, :CN ⁻ , which initiates nucleophilic attack on the delta-positive carbonyl carbon [1]	1
7(b)	Disappearance of the strong carbonyl C=O absorption peak at ~1730 cm ⁻¹ and aldehydic C–H doublet at 2720/2820 cm ⁻¹ as ethanal is consumed [1] Appearance of a sharp nitrile C≡N absorption peak at 2200–2250 cm ⁻¹ [1] Appearance of a broad alcohol O–H absorption band at 3200–3600 cm ⁻¹ as the cyanohydrin forms [1]	3
7(c)	Yes, C2 is a chiral center bonded to four different groups: –H, –CH3, –OH, and –CN [1] Two 3D tetrahedral structures drawn around C2 with wedge, dash, and normal lines [1] Structures are non-superimposable mirror images (enantiomers) [1]	3
8(a)	Elimination (or dehydrohalogenation) [1] Product W: Propene (CH3CH=CH2) [1]	2
8(b)	C=C double bond stretch at 1620–1680 cm ⁻¹ [1] C–H stretch (sp ² C–H) at 3000–3100 cm ⁻¹ [1]	2
8(c)	C–Br bond stretch at 500–600 cm ⁻¹ [1]	1
9(a)	Pentan-2-one is a ketone; pentanal is an aldehyde [1] C=O (carbonyl) bond stretch [1]	2
9(b)	Pentanal shows two characteristic sharp absorption peaks at 2720 cm ⁻¹ and 2820 cm ⁻¹ due to the aldehydic C–H stretch (Fermi resonance doublet) [1] Pentan-2-one has no hydrogen atom directly attached to the carbonyl carbon and lacks these peaks [1]	2
9(c)	Tri-iodomethane (iodoform) test: alkaline aqueous iodine (I ₂ / NaOH(aq)) [1] Pentan-2-one has a CH ₃ C=O group and gives a pale yellow precipitate of CHI ₃ ; pentanal does not [1]	2
10(a)	Absorb IR: Carbon dioxide (CO ₂) and water vapor (H ₂ O) (or methane CH ₄) [1] Do not absorb IR: Nitrogen (N ₂) and oxygen (O ₂) (or argon Ar) [1]	2
10(b)	CO ₂ is linear and non-polar at rest, but asymmetrical stretching and bending vibrations create a temporary changing dipole moment [1] A changing dipole moment allows interaction with and absorption of infrared photons of matching frequency [1]	2
10(c)	Methane absorbs infrared radiation in a wavelength window where CO ₂ and H ₂ O do not absorb strongly [1] The C–H bonds in methane have higher molar absorptivity (intensity of absorption) in the infrared region [1]	2
11(a)	In the gas phase, molecules are widely separated with negligible intermolecular hydrogen bonding; all O–H bonds vibrate freely at a single discrete frequency (3650 cm ⁻¹) [1] In the liquid phase, extensive intermolecular hydrogen bonding occurs between –OH groups [1] Hydrogen bonds exist across a wide continuum of lengths and bond angles, creating a broad distribution of vibrational energy levels and a broadened absorption band (3200–3600 cm ⁻¹) [1]	3
11(b)	Hydrogen bonding pulls electron density away from the covalent O–H bond, weakening the bond [1] A weaker bond has a lower force constant (k), which decreases the vibrational frequency according to Hooke's Law (wavenumber proportional to sqrt(k)) [1]	2
11(c)	No; C–H bonds do not participate significantly in hydrogen bonding, so their force constants and vibrational frequencies remain sharp and unaffected [1]	1
12(a)	A is primary (1°), B is secondary (2°), C is tertiary (3°) [1]	1
12(b)	All three isomers contain identical functional groups: an alcohol O–H bond and sp ³ C–H bonds [1] All three will exhibit the same broad O–H stretch at 3200–3600 cm ⁻¹ and alkyl C–H stretches at 2850–2950 cm ⁻¹ , with no diagnostic carbonyl or unsaturated peaks [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
12(c)	Add acidified K ₂ Cr ₂ O ₇ and warm: tertiary alcohol C resists oxidation, so its IR spectrum remains unchanged (retains 3350 cm ⁻¹ O-H band) [1] Secondary alcohol B oxidizes to ketone butan-2-one: broad O-H disappears, replaced by a single sharp C=O peak at ~1715 cm ⁻¹ with no aldehydic C-H doublet [1] Primary alcohol A oxidizes to butanal/butanoic acid: shows C=O peak with aldehydic doublet (2720/2820 cm ⁻¹) or broad carboxylic acid O-H envelope (2500-3000 cm ⁻¹) [1]	3
13(a)	CH ₃ COOCH ₃ + H ₂ O <=> CH ₃ COOH + CH ₃ OH [1]	1
13(b)	Strong ester C=O stretch at 1735-1750 cm ⁻¹ [1] Strong C-O ester stretch at 1150-1250 cm ⁻¹ (with no absorption above 3000 cm ⁻¹ except alkyl C-H at 2850-2950 cm ⁻¹) [1]	2
13(c)	Broad carboxylic acid O-H envelope at 2500-3000 cm ⁻¹ from ethanoic acid [1] Broad alcohol O-H stretch at 3200-3600 cm ⁻¹ from methanol [1]	2
14(a)	Empirical formula mass = 2(12.0) + 4(1.0) + 16.0 = 44.0; 88 / 44 = 2; Molecular formula = C ₄ H ₈ O ₂ [1]	1
14(b)	1740 cm ⁻¹ is C=O stretch and 1240 cm ⁻¹ is C-O stretch [1] Absence of 2500-3600 cm ⁻¹ band rules out carboxylic acid and alcohol; therefore, D is an ester [1]	2
14(c)	Structural formula: CH ₃ COOCH ₂ CH ₃ [1] Name: Ethyl ethanoate [1]	2
14(d)	CH ₃ COOCH ₂ CH ₃ + NaOH -> CH ₃ COONa + CH ₃ CH ₂ OH [1]	1
15(a)	Steam and concentrated phosphoric acid (H ₃ PO ₄) catalyst [1] Temperature of 300 °C and pressure of 60 atm (6 MPa) [1]	2
15(b)	Disappearance of the alkene C=C absorption at 1620-1680 cm ⁻¹ and alkene =C-H stretch at 3000-3100 cm ⁻¹ [1] Appearance of the strong, broad alcohol O-H stretch at 3200-3600 cm ⁻¹ and C-O stretch at 1050 cm ⁻¹ [1]	2
15(c)	Poly(ethene) is a completely saturated hydrocarbon containing only C-C and C-H sigma bonds [1] Its spectrum displays only sp ³ C-H stretches (2850-2950 cm ⁻¹) and -CH ₂ - bending modes (~1460 cm ⁻¹), lacking any unsaturated C=C bands [1]	2
16(a)	B [1] (Propanoic acid contains a C=O bond at 1715 cm ⁻¹ and a carboxylic acid O-H envelope at 2500-3000 cm ⁻¹)	1
17(a)	C [1] (O ₂ is a homonuclear diatomic molecule with zero dipole moment change during vibration)	1
18(a)	B [1] (C=N nitrile stretch absorbs characteristically at 2200-2250 cm ⁻¹)	1
19(a)	B [1] (Reduction of propan-2-one consumes the C=O group at 1715 cm ⁻¹ and forms propan-2-ol, generating the broad alcohol O-H band at 3200-3600 cm ⁻¹)	1
20(a)	C [1] (Ethanoic acid possesses a carboxylic acid O-H envelope at 2500-3000 cm ⁻¹ ; methyl methanoate is an ester lacking an O-H bond)	1
21(a)	CH ₃ CH ₂ CN + HCl + 2H ₂ O -> CH ₃ CH ₂ COOH + NH ₄ Cl [1]	1
21(b)	Disappearance of the sharp nitrile C=N absorption at 2200-2250 cm ⁻¹ [1] Appearance of the strong carbonyl C=O peak at ~1715 cm ⁻¹ and very broad carboxylic acid O-H envelope at 2500-3000 cm ⁻¹ [1]	2
21(c)	Reduction produces propylamine (CH ₃ CH ₂ CH ₂ NH ₂) [1] The product spectrum exhibits an N-H stretching doublet at 3300-3500 cm ⁻¹ with no carbonyl C=O absorption at 1715 cm ⁻¹ and no carboxylic O-H envelope [1]	2
22(a)	Hex-1-ene exhibits an alkene C=C stretching absorption at 1620-1680 cm ⁻¹ and an sp ² =C-H stretch at 3000-3100 cm ⁻¹ [1] Hexane is an alkane lacking C=C bonds; it exhibits only sp ³ C-H stretches below 3000 cm ⁻¹ (2850-2950 cm ⁻¹) and no absorption between 1620 and 1680 cm ⁻¹ [1]	2
22(b)	Add aqueous bromine (bromine water, Br ₂ (aq)) at room temperature [1] Hex-1-ene rapidly decolourises bromine water from orange/brown to colourless; hexane causes no color change in the dark [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
23(a)	But-1-ene, cis-but-2-ene ((Z)-but-2-ene), and trans-but-2-ene ((E)-but-2-ene) [1] But-1-ene and but-2-ene are positional isomers; cis- and trans-but-2-ene are geometrical (stereoisomers / diastereomers) [1]	2
23(b)	Both geometrical isomers possess identical functional groups, bond types, and atoms [1] Both exhibit sp ² C-H stretches at ~3020 cm ⁻¹ and C=C stretches at ~1660 cm ⁻¹ ; they differ primarily in the fingerprint region (< 1500 cm ⁻¹) due to out-of-plane =C-H bending [1]	2
23(c)	Trans-but-2-ene has a centrosymmetric structure; when the C=C bond stretches symmetrically, there is zero net change in dipole moment [1] Vibrations that produce no change in dipole moment are infrared-inactive (selection rule) [1]	2
24(a)	C-H bond stretch (in the CH ₃ CH ₂ - alkyl group) [1] Water vapor contains O-H bonds that absorb above 3200 cm ⁻¹ and bending modes at ~1600 cm ⁻¹ , having negligible absorption at 2950 cm ⁻¹ [1]	2
24(b)	The absorbance is directly proportional to ethanol concentration (Beer-Lambert Law: A = epsilon * c * l) [1] A fixed partition ratio exists between ethanol in pulmonary alveolar blood and exhaled breath (approx 2100:1), allowing direct calibration [1]	2
25(a)	Peak at 1720 cm ⁻¹ confirms carbonyl C=O group [1] Absence of 3200–3600 cm ⁻¹ rules out alcohol (e.g., prop-2-en-1-ol); absence of 2720/2820 cm ⁻¹ doublet rules out aldehyde (propanal) [1] Therefore, X is the ketone: propan-2-one (CH ₃ COCH ₃) [1]	3
25(b)	Displayed formula of 2-hydroxy-2-methylpropanenitrile, (CH ₃) ₂ C(OH)CN, with all bonds displayed [1]	1
25(c)	Loss of carbonyl C=O peak at 1720 cm ⁻¹ [1] Appearance of sharp nitrile C≡N peak at 2200–2250 cm ⁻¹ and broad alcohol O-H band at 3200–3600 cm ⁻¹ [1]	2
26(a)	Vaporised molecules are bombarded by high-energy electrons (approx 70 eV), knocking an electron out of the molecule to form a radical cation [1] Equation: CH ₃ COCH ₃ + e ⁻ → [CH ₃ COCH ₃] ^{•+} + 2e ⁻ (or M + e ⁻ → M ^{•+} + 2e ⁻) [1]	2
26(b)	It gives the relative molecular mass (M _r) of the compound [1]	1
26(c)	n = (100 * abundance of [M+1] ⁺) / (1.1 * abundance of [M] ⁺) [1]	1
26(d)	n = (100 * 0.95) / (1.1 * 28.0) = 95 / 30.8 [1] n = 3.08 approx 3 carbon atoms [1]	2
27(a)	Two peaks separated by 2 m/z units ([M] ⁺ and [M+2] ⁺) [1] Peak height ratio of [M] ⁺ : [M+2] ⁺ is 3 : 1 (due to 75% ³⁵ Cl and 25% ³⁷ Cl) [1]	2
27(b)	Two peaks separated by 2 m/z units of approximately equal intensity (twin peaks) [1] Peak height ratio of [M] ⁺ : [M+2] ⁺ is 1 : 1 (due to ~50% ⁷⁹ Br and ~50% ⁸¹ Br) [1]	2
27(c)	Binomial expansion (3/4 + 1/4) ² = (9/16) ³⁵ Cl- ³⁵ Cl + (6/16) ³⁵ Cl- ³⁷ Cl + (1/16) ³⁷ Cl- ³⁷ Cl [1] [M] ⁺ : [M+2] ⁺ : [M+4] ⁺ [1] Ratio is 9 : 6 : 1 [1]	3
27(d)	Ratio is 1 : 2 : 1 (from binomial (1/2 + 1/2) ² = 1/4 : 2/4 : 1/4) [1]	1
28(a)	M _r = 79 + 35 + 2(12) + 4(1) = 142 [1]	1
28(b)	m/z = 142 corresponds to [C ₂ H ₄ ⁷⁹ Br ³⁵ Cl] ⁺ [1] m/z = 144 corresponds to [C ₂ H ₄ ⁸¹ Br ³⁵ Cl] ⁺ and [C ₂ H ₄ ⁷⁹ Br ³⁷ Cl] ⁺ ; m/z = 146 corresponds to [C ₂ H ₄ ⁸¹ Br ³⁷ Cl] ⁺ [1]	2
28(c)	Probabilities: ⁷⁹ Br* ³⁵ Cl = (1/2)*(3/4) = 3/8; (⁸¹ Br* ³⁵ Cl + ⁷⁹ Br* ³⁷ Cl) = (1/2)*(3/4) + (1/2)*(1/4) = 4/8; ⁸¹ Br* ³⁷ Cl = (1/2)*(1/4) = 1/8 [1] Ratio at m/z 142 : 144 : 146 is 3 : 4 : 1 [1]	2
29(a)	The most intense (tallest) peak in the mass spectrum, assigned an arbitrary relative abundance of 100% [1]	1
29(b)	Species: [CH ₃ CO] ⁺ (acylium / ethanoyl cation) [1] Equation: [CH ₃ COCH ₃] ^{•+} → [CH ₃ CO] ⁺ + .CH ₃ [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
29(c)	Species at $m/z = 15$: $[\text{CH}_3]^+$ (methyl carbocation) [1] Neutral radicals carry no electric charge; only positively charged ions are accelerated and deflected by magnetic/electric fields to reach the detector [1]	2
30(a)	Moles: $\text{C} = 85.7 / 12.0 = 7.14$; $\text{H} = 14.3 / 1.0 = 14.3$ [1] Ratio: $7.14 : 14.3 = 1 : 2$; Empirical formula = CH_2 [1]	2
30(b)	$n = (100 * 1.70) / (1.1 * 38.5) = 170 / 42.35$ [1] $n = 4.01$ approx 4 carbon atoms [1]	2
30(c)	Molecular formula: C_4H_8 [1] Any two of: but-1-ene, but-2-ene, 2-methylpropene [1]	2
31(a)	Breaking of a carbon-carbon bond adjacent to the carbon atom bearing the -OH group (the $\text{C}_\alpha - \text{C}_\beta$ bond) [1]	1
31(b)	Fragment cation: $[\text{CH}_3\text{CH}_2\text{CH}=\text{OH}]^+$ (or $[\text{C}_3\text{H}_7\text{O}]^+$) at $m/z = 59$ [1] Neutral radical lost: $\cdot\text{CH}_2\text{CH}_3$ (ethyl radical, 29 mass units) [1]	2
31(c)	$m/z = 45$: $[\text{CH}_3\text{CH}=\text{OH}]^+$ (loss of propyl radical $\cdot\text{CH}_2\text{CH}_2\text{CH}_3$, 43 mass units) [1] $m/z = 73$: $[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{OH}]^+$ (loss of methyl radical $\cdot\text{CH}_3$, 15 mass units) [1]	2
32(a)	$[\text{CH}_3\text{COOCH}_3]^+$. (or $[\text{C}_3\text{H}_6\text{O}_2]^+$.) [1]	1
32(b)	$m/z = 43$: $[\text{CH}_3\text{CO}]^+$ (ethanoyl / acylium cation) [1] $m/z = 59$: $[\text{COOCH}_3]^+$ (methoxycarbonyl cation) [1]	2
32(c)	Neutral radical: $\cdot\text{OCH}_3$ (methoxy radical) [1] Equation: $[\text{CH}_3\text{COOCH}_3]^+ \rightarrow [\text{CH}_3\text{CO}]^+ + \cdot\text{OCH}_3$ [1]	2
33(a)	$[\text{C}_2\text{H}_5\text{CO}]^+$ (loss of the aldehydic H. atom from the $\text{C}=\text{O}$ group) [1] Ketones have no hydrogen atom bonded directly to the carbonyl carbon, so loss of H. to form an acylium ion cannot occur [1]	2
33(b)	Propan-2-one cleaves a $\text{C}-\text{CH}_3$ bond adjacent to $\text{C}=\text{O}$, yielding the stable $[\text{CH}_3\text{CO}]^+$ cation at $m/z = 43$ [1] Propanal cleaves the $\text{C}-\text{C}$ bond between C_2 and C_3 , yielding the ethyl cation $[\text{C}_2\text{H}_5]^+$ or formyl cation $[\text{CHO}]^+$ at $m/z = 29$ [1]	2
33(c)	$[\text{CHO}]^+$ (formyl cation) [1] $[\text{C}_2\text{H}_5]^+$ (ethyl carbocation) [1]	2
34(a)	Water molecule, H_2O (mass 18) [1]	1
34(b)	Methyl radical, $\cdot\text{CH}_3$ (mass 15) [1]	1
34(c)	Carbon monoxide (CO) or ethene (C_2H_4) [1]	1
34(d)	($M - 17$): loss of hydroxyl radical ($\cdot\text{OH}$), leaving the acylium cation $[\text{RCO}]^+$ [1] $m/z = 45$: carboxyl cation, $[\text{COOH}]^+$ [1]	2
35(a)	$m/z = 136$ is $[\text{C}_n\text{H}_{2n+1}]^+ 79\text{Br}$; Alkyl group mass = $136 - 79 = 57$ [1] $\text{C}_n\text{H}_{2n+1} = 57$ gives $12n + 2n + 1 = 57 \Rightarrow 14n = 56 \Rightarrow n = 4$; Formula = $\text{C}_4\text{H}_9\text{Br}$ [1]	2
35(b)	$[\text{C}_4\text{H}_9]^+$ (butyl carbocation) [1]	1
35(c)	Structure: $(\text{CH}_3)_3\text{CBr}$ [1] Name: 2-bromo-2-methylpropane [1]	2
35(d)	Tertiary carbocation $(\text{CH}_3)_3\text{C}^+$ is highly stabilized by the positive inductive effect (+I) of three electron-releasing methyl groups [1] Steric hindrance from three bulky methyl groups blocks backside attack by the OH^- nucleophile (ruling out $\text{S}_\text{N}2$) [1]	2
36(a)	C [1] ($n = (100 * 2.20) / (1.1 * 50.0) = 220 / 55.0 = 4$)	1
37(a)	C [1] (Equal peak heights at M and $M+2$ signify the 1:1 abundance ratio of 79Br and 81Br)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
38(a)	B [1] (Alpha-cleavage of propan-2-ol expels a .CH ₃ radical (mass 15), leaving the resonance-stabilized oxonium ion [CH ₃ CH=OH] ⁺ at m/z = 60 - 15 = 45)	1
39(a)	C [1] (Two chlorine atoms give peak ratios corresponding to (3/4 + 1/4) ² = 9 : 6 : 1)	1
40(a)	B [1] (The tropylium ion [C ₇ H ₇] ⁺ is an aromatic, resonance-stabilized 7-membered carbocation with 6 pi electrons at m/z = 91)	1
41(a)	Cleavage of any of the three C-CH ₃ bonds in 2-methylpropane produces a secondary carbocation [(CH ₃) ₂ CH] ⁺ (or 2-propyl cation) [1] Secondary carbocations are much more stable than primary carbocations due to hyperconjugation / positive inductive effect (+I), leading to preferential cleavage [1]	2
41(b)	[CH(CH ₃) ₃] ⁺ → [(CH ₃) ₂ CH] ⁺ + .CH ₃ [1]	1
41(c)	Stability order: tertiary (3°) > secondary (2°) > primary (1°) [1] Alkyl groups are electron-donating by positive inductive effect (+I) / hyperconjugation, dispersing the positive charge over multiple carbons [1]	2
42(a)	Mass C = 2.64 * (12.0 / 44.0) = 0.72 g; Mass H = 1.08 * (2.0 / 18.0) = 0.12 g [1] Mass O = 1.32 - (0.72 + 0.12) = 0.48 g [1] Moles: C = 0.72/12 = 0.06; H = 0.12/1 = 0.12; O = 0.48/16 = 0.03; Ratio = 2 : 4 : 1 => Empirical formula = C ₂ H ₄ O [1]	3
42(b)	n = (100 * 1.85) / (1.1 * 42.0) = 185 / 46.2 = 4.00 carbons [1] Empirical formula mass (C ₂ H ₄ O) = 44; Mr = 88; Molecular formula = C ₄ H ₈ O ₂ [1]	2
42(c)	IR peaks confirm carboxylic acid (-COOH) [1] Structure: CH ₃ CH ₂ CH ₂ COOH (or (CH ₃) ₂ CHCOOH); IUPAC name: Butanoic acid (or 2-methylpropanoic acid) [1]	2
43(a)	Alpha-cleavage breaks the C1-C2 bond, expelling an ethyl radical (.CH ₂ CH ₃ , mass 29) [1] Fragment cation: [CH ₂ =NH ₂] ⁺ (iminium ion) at m/z = 30 [1]	2
43(b)	The nitrogen lone pair donates electron density into the empty p-orbital of the adjacent carbon [1] Resonance stabilization: [H ₂ C-NH ₂] ⁺ <-> [H ₂ C=NH ₂] ⁺ with all atoms having a full octet of valence electrons [1]	2
43(c)	m/z = 44 [1] Fragment ion: [CH ₃ CH=NH ₂] ⁺ (loss of a methyl radical .CH ₃ , mass 15) [1]	2
44(a)	m/z = 43: [CH ₃ CO] ⁺ (loss of propyl radical .CH ₂ CH ₂ CH ₃ , mass 43) [1] m/z = 71: [CH ₃ CH ₂ CH ₂ CO] ⁺ (loss of methyl radical .CH ₃ , mass 15) [1]	2
44(b)	m/z = 57: [CH ₃ CH ₂ CO] ⁺ (propanoyl cation) [1] Loss of ethyl radical (.CH ₂ CH ₃ , mass 29) [1]	2
44(c)	Pentan-2-one gives major peaks at m/z = 43 and 71 with no significant peak at 57; pentan-3-one gives a dominant peak at m/z = 57 [1]	1
45(a)	n = (100 * 0.83) / (1.1 * 15.0) = 83 / 16.5 [1] n = 5.03 approx 5 carbon atoms [1]	2
45(b)	Formula mass: 5(12) + 2(16) = 92; Remaining mass = 102 - 92 = 10 H; Molecular formula = C ₅ H ₁₀ O ₂ [1]	1
45(c)	Structural formula: CH ₃ CH ₂ COOCH ₂ CH ₃ [1] IUPAC name: Ethyl propanoate [1]	2
45(d)	Propanoic acid (CH ₃ CH ₂ COOH) [1] Ethanol (CH ₃ CH ₂ OH) [1]	2
46(a)	Moles: C = 66.7/12 = 5.56; H = 11.1/1 = 11.1; O = 22.2/16 = 1.39; Ratio = 4 : 8 : 1 => Empirical = C ₄ H ₈ O (mass = 72) [1] Since Mr = 72, the molecular formula is C ₄ H ₈ O [1]	2
46(b)	Positive 2,4-DNPH confirms carbonyl group; negative Tollens' rules out aldehyde, confirming ketone [1] Structural formula: CH ₃ COCH ₂ CH ₃ (butan-2-one) [1]	2
46(c)	m/z = 43: [CH ₃ CO] ⁺ (loss of ethyl radical .CH ₂ CH ₃) [1] m/z = 57: [CH ₃ CH ₂ CO] ⁺ (loss of methyl radical .CH ₃) [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
46(d)	$\text{CH}_3\text{COCH}_2\text{CH}_3 + 3\text{I}_2 + 4\text{OH}^- \rightarrow \text{CHI}_3 + \text{CH}_3\text{CH}_2\text{COO}^- + 3\text{I}^- + 3\text{H}_2\text{O}$ [1] CHI_3 is tri-iodomethane (yellow precipitate) [1]	2
47(a)	1-chloropropane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ [1] 2-chloropropane, $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_3$ [1]	2
47(b)	In 2-chloropropane, loss of a .Cl radical gives the secondary propyl carbocation $[(\text{CH}_3)_2\text{CH}]^+$ at $m/z = 43$, which is highly stable and forms the base peak [1] In 1-chloropropane, loss of a methyl radical .CH ₃ produces the chloromethyl cation $[\text{CH}_2\text{CH}_2\text{Cl}]^+$ at $m/z = 63/65$ (retaining the chlorine isotope pattern) [1] Therefore, isomer X is 2-chloropropane and isomer Y is 1-chloropropane [1]	3
47(c)	$m/z = 43$: $[\text{C}_3\text{H}_7]^+$ (or $[(\text{CH}_3)_2\text{CH}]^+$) [1] $m/z = 63$: $[\text{C}_2\text{H}_4\text{ }^{35}\text{Cl}]^+$ [1]	2
48(a)	Skeletal structures of propan-1-ol, propan-2-ol, and methoxyethane [1] Any two correct skeletal structures [1]	2
48(b)	Propan-1-ol and propan-2-ol exhibit a strong, broad O-H stretch at 3200–3600 cm^{-1} [1] Methoxyethane is an ether with no O-H bond, showing no absorption above 3000 cm^{-1} except alkyl C-H stretches [1]	2
48(c)	Isomer 1 is propan-1-ol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \rightarrow [\text{CH}_2\text{OH}]^+ m/z 31 + .\text{C}_2\text{H}_5$) [1] Isomer 2 is propan-2-ol ($\text{CH}_3\text{CH}(\text{OH})\text{CH}_3 \rightarrow [\text{CH}_3\text{CHOH}]^+ m/z 45 + .\text{CH}_3$) [1]	2
48(d)	Methoxyethane (ethyl methyl ether) [1]	1
49(a)	Moles: $\text{C} = 40.0/12 = 3.33$; $\text{H} = 6.7/1 = 6.7$; $\text{O} = 53.3/16 = 3.33$; Ratio = 1 : 2 : 1 $\Rightarrow \text{CH}_2\text{O}$ [1] Empirical formula mass = 30.0 [1]	2
49(b)	$n = (100 \times 0.55) / (1.1 \times 25.0) = 55 / 27.5 = 2.00$ carbon atoms [1] Empirical formula $\text{CH}_2\text{O} \times 2$ gives molecular formula $\text{C}_2\text{H}_4\text{O}_2$ ($M_r = 60$) [1]	2
49(c)	Acidic pH and 2500–3000 cm^{-1} broad band confirm carboxylic acid [1] Structural formula: CH_3COOH (ethanoic acid) [1]	2
49(d)	CH_3COCl (ethanoyl chloride) [1] $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl} + \text{SO}_2 + \text{HCl}$ [1]	2
50(a)	Carbonyl group (aldehyde or ketone / $\text{C}=\text{O}$) [1]	1
50(b)	Positive Tollens' and 2720/2820 cm^{-1} doublet confirm aldehyde [1] Base peak at $m/z = 57$ is the butyl cation $[\text{C}_4\text{H}_9]^+$ (or propionyl $[\text{CH}_3\text{CH}_2\text{CO}]^+$ formed by alpha-cleavage of pentanal) [1] Displayed formula of pentanal (or 2-methylbutanal); IUPAC name: Pentanal (or 2-methylbutanal) [1]	3
50(c)	Methyl ketone group, $\text{CH}_3-\text{C}=\text{O}$ (or methyl carbonyl group attached to carbon) [1]	1
50(d)	Displayed formula of pentan-2-one ($\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$) [1] Systematic name: Pentan-2-one [1]	2
50(e)	Pentan-2-ol [1] Carbonyl peak at 1715 cm^{-1} completely disappears and is replaced by a strong, broad alcohol O-H band at 3200–3600 cm^{-1} [1]	2